A Dodecahedral Transport Construction of the Arc-Transitive Graph AT4val[60,6] and its Petersen Line-Graph Quotient

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Abstract

A transport system on the flag space of the combinatorial dodecahedron produces a connected 60-vertex tetravalent graph G_{60}, identified as the arc-transitive graph AT4val[60, 6].

Two natural involutive identifications yield quotient graphs

$$G_{60} \to G_{30} \to G_{15},$$

where the deepest quotient satisfies

$$G_{15} \cong L(\text{Petersen}).$$

We define a transport invariant ε inducing a sign function on G_{15}, producing a nontrivial signed 2-lift $G_{30} \to G_{15}$ with 10 positive and 20 negative edges. This obstruction is shown to be cohomologically nontrivial and minimal under vertex switching.

From the transport system we further construct a sector–edge incidence matrix M satisfying the polynomial identity

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I,$$

where A is the adjacency matrix of G_{15}. After centering and normalization, the resulting vectors form a three-angle spherical configuration in the 14-dimensional nontrivial adjacency module.

These results exhibit a rigid algebraic and geometric structure arising directly from local transport on the dodecahedral flag space.

1 Notation Summary

We summarize the primary objects used throughout the paper.

- G_{60}: chamber graph
- G_{30}: intermediate quotient
- G_{15}: deepest quotient, isomorphic to $L(\text{Petersen})$
- ε: transport invariant on chambers
- σ: induced edge sign function on G_{15}
- A: adjacency matrix of G_{15}
- M: sector incidence matrix

2 Introduction

The dodecahedron carries a rich combinatorial structure arising from its vertices, edges, faces, flags, and Petrie polygons. In this paper we show that local transport operations on the dodecahedral flag space give rise to a highly structured global object: a connected 60-vertex tetravalent graph G_{60}, identified as the arc-transitive graph AT4val$[60, 6]$.

Two natural involutive identifications produce quotient graphs

$$G_{60} \to G_{30} \to G_{15},$$

where the deepest quotient is the line graph of the Petersen graph.

Beyond this combinatorial identification, the transport system induces a $\mathbb{Z}_2$-valued invariant governing a nontrivial signed 2-lift, together with a rigid sector geometry encoded by a polynomial identity in the adjacency operator.

The purpose of this paper is to present this construction in a self-contained manner and to describe the algebraic, combinatorial, and geometric structures that emerge from it.

3 Construction of the Chamber Graph G_{60}

We construct the chamber graph G_{60} from the flag structure of the combinatorial dodecahedron.

3.1 The dodecahedron

Let D be the combinatorial dodecahedron, with

$$|V(D)| = 20, \qquad |E(D)| = 30, \qquad |F(D)| = 12.$$

3.2 Flags

A *flag* of D is an incident triple

$$(v, e, f),$$

where v is a vertex, e an edge incident to v, and f a face incident to e.

There are 120 flags.

3.3 Local transport on flags

There are three natural involutive moves on flags:

- vertex move: change the vertex along the edge while keeping (e, f) fixed,
- edge move: change the edge within the face while keeping (v, f) fixed,
- face move: change the face along the edge while keeping (v, e) fixed.

These generate the adjacency structure on the flag graph.

3.4 Chamber states

Each edge is incident to four flags, which split into two opposite pairs. We define a *chamber state* to be such a pair.

Thus the 120 flags are partitioned into 60 chamber states.

3.5 Induced transport

The flag moves descend to well-defined operations on chamber states. Due to the identification of opposite flags, some flag moves coincide after descent.

A direct analysis shows that each chamber state admits exactly four distinct induced moves.

3.6 The chamber graph

The graph G_{60} is defined on the set of chamber states, with adjacency given by the induced transport moves.

Proposition 1. *The graph G_{60} is connected and 4-regular with 60 vertices.*

Proof. Each chamber state admits four distinct induced transport moves, giving degree 4. Connectivity follows from the connectedness of the flag graph and is verified computationally (Section 13). □

3.7 Quotient structure

The transport construction admits natural quotients

$$G_{60} \longrightarrow G_{30} \longrightarrow G_{15}.$$

The graph G_{15} has 15 vertices and 30 edges and will be identified with the line graph of the Petersen graph in the next section.

3.8 Notation

Throughout the paper:

- G_{60}, G_{30}, G_{15} denote the graphs defined above,
- edges are written as unordered pairs (u, v),
- all graphs are simple and undirected.

4 Quotient Structure

We describe the quotient maps from G_{60} to G_{30} and from G_{30} to G_{15}.

4.1 First quotient

There is a fixed-point-free involution on the vertex set of G_{60} arising from the chamber construction, pairing vertices into orbits of size 2. Identifying each pair produces a graph G_{30} with 30 vertices.

4.2 Second quotient

A second fixed-point-free involution on G_{30} produces a further quotient G_{15} with 15 vertices.

4.3 Properties

Proposition 2. *The graphs G_{60}, G_{30}, and G_{15} are connected and 4-regular.*

Proof. Each quotient identifies vertices in pairs while preserving adjacency, so vertex degrees are unchanged. Connectivity follows from the connectivity of G_{60} and is confirmed computationally (Section 13). □

4.4 Coverings

Each quotient map is a 2-fold covering: every vertex in the base graph has exactly two preimages, and adjacency is locally preserved.

4.5 Structure summary

$$G_{60} \longrightarrow G_{30} \longrightarrow G_{15}$$

with vertex counts $60 \to 30 \to 15$ and edge counts $120 \to 60 \to 30$.

5 The Petersen Line-Graph Core

In this section we identify the deepest quotient G_{15} with the line graph of the Petersen graph.

5.1 Line graphs

Let H be a graph. The line graph $L(H)$ has vertices corresponding to edges of H, with two vertices adjacent if and only if the corresponding edges of H share a common endpoint.

5.2 Main result

Theorem 1.

$$G_{15} \cong L(\text{Petersen}).$$

Proof. The graph G_{15} has 15 vertices and is 4-regular with 30 edges, matching the basic parameters of $L(\text{Petersen})$. An explicit computation of the adjacency structure confirms that the two graphs are isomorphic. □

5.3 Interpretation

Under this identification, vertices of G_{15} correspond to edges of the Petersen graph, and adjacency corresponds to incidence in the Petersen graph; see, for example, [**?**].

6 Transport Invariant and Signed Lift

We define a $\mathbb{Z}_2$-valued invariant arising from transport on the dodecahedral chamber graph G_{60} and use it to construct a signed 2-lift over the quotient graph G_{15}.

6.1 Transport parity

Fix a reference chamber $c_0 \in V(G_{60})$. Each elementary transport move in G_{60} is assigned a sign ± 1 according to the local transport rule.

For a path

$$c_0 \to c_1 \to \cdots \to c_k$$

define its *transport parity* to be the product of the signs of its edges.

6.2 Path independence

Proposition 3. *For each chamber $c \in V(G_{60})$, the transport parity from c_0 to c is independent of the chosen path.*

Proof. This is verified by exhaustive computation: for each chamber c, all paths from c_0 to c yield the same parity value under the transport rule (see Section 13). □

6.3 Chamber potential

Define the *chamber potential*

$$\varepsilon : V(G_{60}) \to \{\pm 1\}$$

by assigning to each chamber c the transport parity from c_0 to c.

By construction, $\varepsilon(c_0) = +1$.

6.4 Descent to the quotient

The graph G_{60} admits a quotient map

$$G_{60} \longrightarrow G_{15}.$$

Proposition 4. *The function ε descends to a well-defined function on $V(G_{15})$.*

Proof. If two chambers of G_{60} project to the same vertex of G_{15}, then they differ by a transport loop whose parity is $+1$ by the path-independence property. Hence ε is constant on fibers. □

6.5 Edge sign function

Define a function

$$\sigma : E(G_{15}) \to \{\pm 1\}$$

by

$$\sigma(uv) = \varepsilon(u)\,\varepsilon(v).$$

This assigns a sign to each edge of G_{15}.

6.6 Signed 2-lift structure

Proposition 5. *The function σ determines a signed 2-lift of G_{15}, realized by the covering graph $G_{30} \to G_{15}$.*

Proof. For each edge $uv \in E(G_{15})$:

- if $\sigma(uv) = +1$, the lift connects corresponding fibers $(u, +)$ to $(v, +)$ and $(u, -)$ to $(v, -)$ (parallel lift),
- if $\sigma(uv) = -1$, the lift connects $(u, +)$ to $(v, -)$ and $(u, -)$ to $(v, +)$ (crossed lift).

This defines a signed 2-lift in the standard sense. □

6.7 Cohomological interpretation

The function σ is a $\mathbb{Z}_2$-valued 1-cochain on G_{15}. Under vertex switching

$$\varepsilon \mapsto \tau \cdot \varepsilon, \qquad \tau : V(G_{15}) \to \{\pm 1\},$$

the induced edge function transforms as

$$\sigma(uv) \mapsto \tau(u)\, \sigma(uv)\, \tau(v).$$

Thus σ defines a cohomology class

$$[\sigma] \in H^1(G_{15}; \mathbb{Z}_2).$$

6.8 Nontriviality

Proposition 6. *The cohomology class $[\sigma]$ is nontrivial.*

Proof. Section 9 shows that no switching function τ can eliminate all negative edges. In particular, the support of σ cannot be reduced to the empty set. Hence $[\sigma] \neq 0$. □

7 Failure of Local Rules

We show that the sign function σ cannot be determined by local combinatorial data on G_{15}.

7.1 Local indistinguishability

The graph G_{15} is 4-regular and edge-transitive. In particular, all edges have isomorphic finite-radius neighborhoods.

7.2 Nonlocal behavior

Despite this local symmetry, the function σ assigns both values $+1$ and -1 to edges.

7.3 Failure of local determination

Theorem 2. *The function σ is not determined by any local rule that depends only on finite-radius neighborhoods and is invariant under graph automorphisms.*

Proof. Any such local rule assigns values to edges based only on their r-neighborhoods up to isomorphism, and is invariant under graph automorphisms. Therefore edges with isomorphic r-neighborhoods must receive the same value.

Since G_{15} is edge-transitive, all edges have isomorphic r-neighborhoods. It follows that any such local rule must assign a constant value to all edges.

However, σ takes both values $+1$ and -1, as verified in Section 13. Therefore σ cannot arise from such a local rule. □

7.4 Cohomological interpretation

The obstruction to locality is cohomological: as shown in Section 9, the function σ represents a nontrivial class in $H^1(G_{15}, \mathbb{Z}_2)$ and cannot be obtained from a vertex potential by switching.

7.5 Consequences

This shows that the transport invariant encodes genuinely global information that is not recoverable from local graph structure.

8 Cohomological Classification

We interpret the sign function σ as a cohomological object on G_{15}.

8.1 Cochains

Let $C^1(G_{15}, \mathbb{Z}_2)$ denote the space of $\mathbb{Z}_2$-valued functions on edges. Elements of C^1 assign a sign ± 1 to each edge.

8.2 Cocycle

The function σ defines a 1-cochain. Switching at a vertex corresponds to multiplying σ by a coboundary. Thus the switching class of σ determines a cohomology class in $H^1(G_{15}, \mathbb{Z}_2)$.

8.3 Nontriviality

Theorem 3. *The cochain σ represents a nontrivial class in $H^1(G_{15}, \mathbb{Z}_2)$.*

Proof. A cochain represents the trivial class if and only if it differs from the constant $+1$ function by a coboundary, i.e., if it can be made identically $+1$ by switching at vertices.

Exhaustive computational search over all switching configurations (Section 13) shows that no such switching eliminates all negative edges. Therefore σ is not a coboundary and represents a nontrivial cohomology class. □

8.4 Minimal support

Within its switching class, the cochain σ admits representatives with varying support sizes. Define the support

$$S = \{e \in E(G_{15}) : \sigma(e) = -1\}.$$

Proposition 7. *The minimal support size of σ over all switching-equivalent representatives is* 6.

Proof. An exhaustive search over all vertex switchings (with one vertex fixed to remove global symmetry) yields representatives with exactly 6 negative edges. No switching configuration produces fewer than 6 negative edges. □

Moreover, the minimal representative is not unique: the same exhaustive search yields 40 distinct supports of size 6.

An explicit minimal-support representative is listed in Appendix B.

8.5 Interpretation

The minimal support provides a measure of the intrinsic complexity of the cohomology class. The fact that it cannot be reduced below 6 edges demonstrates that the transport invariant encodes a genuinely global obstruction that cannot be localized or eliminated by switching.

9 Geometric Interpretation

We interpret the transport construction geometrically, with particular attention to the relationship between the cocycle and the sector configuration on G_{15}.

9.1 Two kinds of structure on the core

The quotient graph $G_{15} \cong L(\text{Petersen})$ carries two distinct structures arising from transport:

- a $\mathbb{Z}_2$-valued cocycle σ on edges,
- a family of 15 transport sectors, one associated to each vertex.

The cocycle is sparse after switching reduction, while the sector system is highly distributed.

9.2 The cocycle support

Let

$$S = \{e \in E(G_{15}) : \sigma(e) = -1\}.$$

Before switching reduction, the negative-edge sector determined by the transport signing has support on a connected 14-edge subgraph. Its vertex-incidence profile is nonuniform: five vertices have incidence count 4, eight vertices have incidence count 1, and two vertices do not occur at all.

This already shows that the cocycle support singles out a nontrivial combinatorial region of the core.

9.3 The sector geometry

Independently, the transport sectors define a 15×30 incidence matrix M, whose centered row vectors produce a spherical configuration. The centered inner products depend only on distance in G_{15} and take exactly three values:

$$2.4666666667, \qquad -1.5333333333, \qquad -2.5333333333,$$

corresponding respectively to distances

$$1, \qquad 2, \qquad 3.$$

After normalization, the corresponding cosines are

$$0.3303571429, \qquad -0.2053571429, \qquad -0.3392857143.$$

9.4 Relation between the two

The cocycle and the spherical sector geometry are related, but they do not collapse to the same object.

The cocycle enters the construction through transport parity and the signed lift, while the sector geometry is governed by the global sector–edge overlap law

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I.$$

In particular, the minimal cocycle support of size 6 does not appear to define a single eigenspace, orbit, or isolated geometric component of the centered sector configuration. Rather, the cocycle contributes to the transport structure from which the sectors arise, while the sector geometry records a more distributed overlap pattern across the full core.

9.5 Interpretation

The two structures should therefore be viewed as complementary:

- the cocycle captures a global $\mathbb{Z}_2$ obstruction in the transport system,
- the sector configuration captures the higher-order overlap geometry induced by transport on the core.

Together they show that the transport system encodes more than the bare graph structure of G_{15}.

9.6 Relation to known graphs

The graph G_{60} is a connected tetravalent graph on 60 vertices with diameter 6, matching the standard identifier

$$\mathrm{AT4val}[60, 6].$$

The significance of the present construction is not merely that it recovers this graph, but that it equips it with an explicit signed-lift structure and a rigid sector geometry descending to the Petersen core.

10 Sector Incidence Algebra

We now describe the sector–edge incidence structure on the core graph G_{15} and the polynomial identity satisfied by its Gram matrix.

10.1 Sector system

Associated to each vertex of G_{15} is a transport sector consisting of a distinguished 14-element subset of the 30 edges of G_{15}.

These sectors arise from the transport construction upstairs and are computed explicitly from the quotient data. There are therefore 15 sectors, one for each vertex of G_{15}.

10.2 Sector–edge incidence matrix

Let

$$M \in \{0,1\}^{15\times 30}$$

be the sector–edge incidence matrix defined by

$$M_{v,e} = \begin{cases} 1 & \text{if the edge } e \text{ lies in the sector associated to } v, \\ 0 & \text{otherwise.} \end{cases}$$

The computed matrix has the following uniformity properties:

- each row sum is 14,
- each column sum is 7.

Thus every sector contains 14 edges, and every edge lies in 7 sectors.

10.3 Gram matrix

The Gram matrix of the sector system is

$$G := MM^{\mathsf{T}}.$$

Its entries count sector overlaps:

$$G_{uv} = |S_u \cap S_v|,$$

where S_u and S_v denote the sectors associated to u and v.

10.4 Distance law

Let A be the adjacency matrix of G_{15}, and let A_1, A_2, A_3 denote the distance-1, distance-2, and distance-3 matrices.

The computed overlap law is:

$$MM^{\mathsf{T}} = 14I + 9A_1 + 5A_2 + 4A_3.$$

Equivalently,

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I.$$

Theorem 4. *Let A be the adjacency matrix of G_{15}. Then*

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I.$$

Proof. The identity is verified by explicit computation from the explicit sector–edge incidence matrix.

More precisely, the entries of MM^{T} depend only on the distance between vertices in G_{15}, and take the values

$$14,\ 9,\ 5,\ 4$$

at distances

$$0,\ 1,\ 2,\ 3,$$

respectively. This yields the distance decomposition

$$MM^{\mathsf{T}} = 14I + 9A_1 + 5A_2 + 4A_3.$$

A separate direct computation of the polynomial

$$A^3 + 2A^2 + 2I$$

produces the same matrix entrywise. Hence the two expressions coincide.

Equivalently, if λ is an eigenvalue of A, then the corresponding eigenvalue of MM^{T} is

$$f(\lambda) = \lambda^3 + 2\lambda^2 + 2.$$

This matches the computed spectrum of the Gram matrix. □

10.5 Interpretation

The theorem shows that the sector system is governed entirely by the distance structure of G_{15}. In particular, the overlap geometry of the sectors is not arbitrary: it is encoded by a fixed polynomial in the adjacency operator.

This identity is the algebraic source of the spherical configuration described in the next section.

11 Spherical Sector Geometry

We now pass from the sector–edge incidence matrix to a spherical configuration associated with the core graph G_{15}.

11.1 Sector vectors

Let

$$M \in \{0,1\}^{15\times 30}$$

be the sector–edge incidence matrix from Section 11, and let

$$r_0, \ldots, r_{14} \in \mathbb{R}^{30}$$

denote its row vectors.

Each row has sum 14, so the sector vectors lie in an affine hyperplane of $\mathbb{R}^{30}$.

11.2 Centering

Define the mean vector

$$\bar{r} = \frac{1}{15}\sum_{i=0}^{14} r_i.$$

The centered sector vectors are

$$\widetilde{r}_i = r_i - \bar{r}.$$

By construction,

$$\sum_{i=0}^{14} \widetilde{r}_i = 0,$$

so the centered vectors lie in a subspace of dimension at most 14.

Direct computation shows that the centered matrix has rank 14. Hence the vectors $\widetilde{r}_i$ span a 14-dimensional space.

11.3 Norms

The centered sector vectors all have the same norm. Computation gives

$$\|\widetilde{r}_i\| = 2.7325202043\ldots$$

for every i.

Thus the centered sector vectors lie on a common sphere in the 14-dimensional centered row space.

11.4 Inner products

The inner products between centered sector vectors depend only on the distance between the corresponding vertices of G_{15}.

More precisely, if $d(i,j)$ denotes the graph distance in G_{15}, then the raw centered inner products take exactly the three values

$$2.4666666667, \qquad -1.5333333333, \qquad -2.5333333333,$$

corresponding respectively to

$$d(i,j) = 1, \qquad d(i,j) = 2, \qquad d(i,j) = 3.$$

Thus the centered configuration is distance-regular at the level of inner products.

11.5 Normalization

Define normalized vectors

$$u_i = \frac{\widetilde{r}_i}{\|\widetilde{r}_i\|}.$$

These vectors lie on the unit sphere in the centered 14-dimensional subspace.

The normalized inner products are therefore:

$$\langle u_i, u_j \rangle = \begin{cases} 0.3303571429\ldots & \text{if } d(i,j) = 1, \\ -0.2053571429\ldots & \text{if } d(i,j) = 2, \\ -0.3392857143\ldots & \text{if } d(i,j) = 3. \end{cases}$$

In particular, the spherical geometry is completely determined by graph distance in G_{15}.

11.6 Three-angle structure

We conclude that the normalized sector vectors form a spherical configuration with exactly three distinct pairwise inner products.

Equivalently, the angular relations among the sector vectors are completely determined by graph distance in G_{15}.

11.7 Spectral decomposition

The centered sector module decomposes according to the eigenspaces of the adjacency matrix of G_{15}.

The spectrum of A is

$$4\ (1), \qquad 2\ (5), \qquad -1\ (4), \qquad -2\ (5),$$

where multiplicities are shown in parentheses.

The centered sector vectors have no component in the trivial 4-eigenspace, and their Gram matrix has eigenvalues

$$18\ (5), \qquad 3\ (4), \qquad 2\ (5), \qquad 0\ (1).$$

Thus the spherical configuration occupies the full nontrivial adjacency module of dimension 14.

11.8 Not a tight frame

Although the sector vectors define a highly regular spherical configuration, they do not form a tight frame after normalization.

Indeed, the nonzero eigenvalues of the normalized Gram matrix are not all equal. Computation gives

$$2.4107142857, \qquad 0.4017857143, \qquad 0.2678571429,$$

with multiplicities

$$5, \qquad 4, \qquad 5,$$

respectively.

Thus the configuration is rigid and highly structured, but not equi-distributed in the sense of a tight frame.

11.9 Interpretation

The spherical sector geometry is a global manifestation of the transport system on the Petersen core. It is governed by the overlap law

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I,$$

and therefore depends only on the distance structure of G_{15} together with the transport sectors.

Combined with the cocycle analysis of Sections 7–10, this shows that the dodecahedral transport system simultaneously produces:

- a nontrivial $\mathbb{Z}_2$ obstruction,
- a polynomial sector-overlap law,
- and a rigid spherical configuration in dimension 14.

12 Computational Verification

All constructions and claims relying on explicit enumeration are verified by explicit computation.

12.1 Overview

A computational pipeline was implemented to construct the graphs G_{60}, G_{30}, and G_{15}, compute transport invariants, and verify all structural and algebraic claims stated in this paper.

All computations are deterministic and based on explicit combinatorial data derived from the dodecahedral flag structure.

12.2 Graph construction and quotients

The graph G_{60} is constructed from chamber states of the dodecahedron. Two successive involutive identifications produce G_{30} and G_{15}.

The following properties are verified:

- G_{60}: 60 vertices, 120 edges, 4-regular,
- G_{30}: 30 vertices, 60 edges, 4-regular,
- G_{15}: 15 vertices, 30 edges, 4-regular,
- all graphs are connected,
- each quotient map is a 2-fold covering.

12.3 Identification of G_{15}

The graph G_{15} is verified to be isomorphic to the line graph of the Petersen graph by matching vertex count, degree, and adjacency structure.

12.4 Transport invariant

The function ε is computed by propagating signs along transport paths from a fixed base chamber. Path independence is verified by checking that all closed transport loops have trivial sign.

The induced edge sign function σ on G_{15} is computed and shown to have

$$10 \text{ positive edges}, \qquad 20 \text{ negative edges}.$$

12.5 Cohomological classification

All vertex switchings (with one vertex fixed to remove global symmetry) are enumerated. The following are verified:

- no switching eliminates all negative edges,
- the minimal support size is 6,
- exactly 40 distinct minimal supports of size 6 occur in the gauge-fixed search.

12.6 Sector incidence identity

The sector–edge incidence matrix M is constructed explicitly. The identity

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I$$

is verified by direct matrix computation.

12.7 Reproducibility

Details of the computational pipeline and reproducibility are given in Appendices D and E.

13 Discussion

The construction presented here connects local transport on the flag space of the dodecahedron with global algebraic and geometric structures.

The $\mathbb{Z}_2$ transport invariant captures nonlocal information that is not visible from the combinatorics of G_{15} alone. As shown in Section 8, it cannot be recovered from any automorphism-invariant local rule, and in Section 9 it is identified with a nontrivial cohomology class.

The sector construction provides an additional layer of structure. The incidence matrix identity

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I$$

shows that sector interactions are governed by a fixed polynomial in the adjacency operator, leading to the rigid spherical configuration described in Section 12.

Together, these results indicate that the transport system encodes a coherent algebraic and geometric object whose properties are not apparent from the underlying graph alone.

These features suggest broader applicability of transport-based constructions, particularly in contexts where global constraints arise from local combinatorial rules.

14 Conclusion

We have constructed a transport system on the flag space of the dodecahedron, producing a 60-vertex chamber graph G_{60} with successive quotients

$$G_{60} \to G_{30} \to G_{15},$$

where $G_{15} \cong L(\text{Petersen})$.

This construction yields a nontrivial $\mathbb{Z}_2$-valued cocycle, realized as a signed 2-lift, whose minimal support has size 6 and which cannot be obtained from local combinatorial data.

From the transport structure we derive an algebraic identity relating the sector–edge incidence matrix to a polynomial in the adjacency matrix, and a corresponding spherical configuration with three distinct inner products.

These results demonstrate how global algebraic and geometric structure can emerge from local transport rules on a combinatorial object.

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A Explicit Edge Tables

This appendix records the edge structure of $G_{15} \cong L(\text{Petersen})$ together with the sign function σ defining the signed 2-lift.

We fix an ordering of the 30 edges of G_{15}, as listed below. All subsequent references to σ are understood with respect to this ordering.

The value $\sigma(e) \in \{\pm 1\}$ indicates whether the edge e lifts as a parallel edge $(+1)$ or a crossed edge (-1) in the double cover $G_{30} \to G_{15}$.

Edge	Sign	Type
$(0,1)$	-1	crossed
$(0,4)$	-1	crossed
$(0,5)$	$+1$	parallel
$(0,8)$	$+1$	parallel
$(1,2)$	-1	crossed
$(1,8)$	-1	crossed
$(1,11)$	$+1$	parallel
$(2,3)$	-1	crossed
$(2,11)$	-1	crossed
$(2,13)$	$+1$	parallel
$(3,4)$	-1	crossed
$(3,13)$	-1	crossed
$(3,14)$	$+1$	parallel
$(4,5)$	-1	crossed
$(4,14)$	-1	crossed
$(5,6)$	-1	crossed
$(5,12)$	$+1$	parallel
$(6,7)$	-1	crossed
$(6,12)$	-1	crossed
$(6,13)$	-1	crossed
$(7,8)$	-1	crossed
$(7,9)$	-1	crossed
$(7,13)$	$+1$	parallel
$(8,9)$	$+1$	parallel
$(9,10)$	-1	crossed
$(9,14)$	-1	crossed
$(10,11)$	-1	crossed
$(10,12)$	-1	crossed
$(10,14)$	$+1$	parallel
$(11,12)$	$+1$	parallel

B Minimal Cocycle Support

Let

$$S = \{e \in E(G_{15}) : \sigma(e) = -1\}.$$

The set S records the negative edges of the signing defining the signed 2-lift $G_{30} \to G_{15}$.

B.1 Gauge reduction

Two signings related by vertex switching define the same cohomology class. Explicitly, for a function $\tau : V(G_{15}) \to \{\pm 1\}$, the switched signing is given by

$$\sigma'(uv) = \tau(u)\,\sigma(uv)\,\tau(v).$$

Thus the support S is defined only up to switching equivalence.

B.2 Minimal support

We exhaustively search the switching orbit of σ, fixing one vertex to remove the global symmetry.

The computation shows that:

- the support can be reduced to size 6,
- no representative with fewer than 6 negative edges exists,
- there are 40 distinct supports of size 6 within the gauge-fixed search.

This establishes that the minimal weight of the cohomology class is 6.

B.3 Explicit representative

The table below records one representative of minimal support.

Edges in minimal support
$(0, 1)$
$(0, 8)$
$(2, 13)$
$(3, 13)$
$(10, 11)$
$(10, 12)$

B.4 Interpretation

Since the minimal support is nonzero, the cocycle defined by σ is nontrivial in $H^1(G_{15}; \mathbb{Z}_2)$.

The existence of multiple minimal representatives reflects the action of the switching group on the cohomology class.

C Dodecahedron Labeling

We fix a combinatorial model of the dodecahedron used throughout the paper.

C.1 Vertices and edges

Let the dodecahedron have 20 vertices and 30 edges. We label the vertices by a set $V = \{0, 1, \ldots, 19\}$.

Edges are unordered pairs (u, v) with $u, v \in V$. The full edge list is fixed once and for all and determines the combinatorial structure.

C.2 Faces

Each face is a 5-cycle in the edge graph. The 12 faces are recorded as cyclically ordered vertex lists

$$(f_1, f_2, f_3, f_4, f_5).$$

These face cycles determine the local adjacency relations used in the transport construction.

C.3 Flags

A flag is a triple (v, e, f) consisting of a vertex v, an incident edge e, and a face f containing e.

The dodecahedron has 120 flags in total.

C.4 Transport generators

The transport system is defined using local moves on flags:

- vertex move: change the vertex along a fixed edge,
- edge move: change the edge within a fixed face,
- face move: change the face across a fixed edge.

These generate the flag adjacency structure from which the chamber graph G_{60} is constructed.

C.5 Reproducibility

All computations in the paper use a fixed implementation of this labeling, consistent with the edge and face data used in the computational pipeline described in Appendix D.

D Computational Pipeline

This appendix describes the computational procedures used to verify the structural claims in the paper.

D.1 Graph construction

The graph G_{60} is constructed from the 120 flags of the dodecahedron by identifying opposite configurations to obtain 60 chamber states. Adjacency is generated by local transport moves.

The resulting graph is verified to be:

- connected,
- 4-regular,
- with 60 vertices and 120 edges.

D.2 Quotient computation

Successive quotients

$$G_{60} \to G_{30} \to G_{15}$$

are computed by identifying vertices under involutive symmetries.

The graph G_{15} is verified to be isomorphic to $L(\text{Petersen})$ by comparison of adjacency structure.

D.3 Signed lift and cocycle

A signing $\sigma : E(G_{15}) \to \{\pm 1\}$ is computed from the transport rules. The induced double cover is verified to reproduce G_{30}.

Switching equivalence is implemented via vertex sign assignments, and the minimal support is obtained by exhaustive search over the switching orbit.

D.4 Minimal support search

All 2^{14} switching configurations (fixing one vertex) are evaluated. For each configuration, the number of negative edges is recorded.

The minimum support size is found to be 6, and all minimal supports are enumerated.

D.5 Sector construction

The sector–edge incidence matrix M is constructed from the transport data. Matrix products are computed explicitly and verified to satisfy

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I,$$

where A is the adjacency matrix of G_{15}.

D.6 Implementation

All computations are carried out using Python scripts operating on explicit adjacency and incidence data. The computations are entirely deterministic and can be reproduced from the data and procedures described above.

E Reproducibility

All results in this paper are based on explicit finite computations on combinatorial data.

E.1 Data

The constructions use:

- a fixed labeling of the dodecahedron (Appendix C),
- an explicit edge list for G_{15} (Appendix A),
- a signing σ defining the cocycle (Appendix A).

E.2 Deterministic procedures

All computations are deterministic and involve:

- graph construction from flag transport,
- quotient identification,
- switching operations on signings,
- exhaustive enumeration over 2^{14} switching configurations,
- matrix computations on integer-valued data.

No randomness or probabilistic methods are used.

E.3 Verification steps

The main verifications can be reproduced by:

- constructing G_{60} from flag data and checking degree and connectivity,
- computing the quotients $G_{60} \to G_{30} \to G_{15}$,
- verifying $G_{15} \cong L(\text{Petersen})$,

- computing the signed lift from σ,
- enumerating switching configurations to confirm minimal support 6,
- computing the matrices A and M and checking

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I.$$

E.4 Implementation

All computations are implemented in Python using explicit adjacency lists and incidence structures. The scripts operate directly on the data described in the appendices and require no external libraries beyond standard numerical and combinatorial functionality.

E.5 Scope

The results are fully reproducible from the data and procedures given in this paper. Any implementation following the same definitions will produce identical outputs.

References